

Supplementary File 13. Evidence Gap Matrix and Research Priorities

Purpose

This supplementary file presents the evidence gaps identified during data charting, domain coding, and cross-domain synthesis. It translates the mapped evidence into a structured gap matrix and research-priority agenda for future work on malaria and sickle cell disease in sub-Saharan Africa.

The gap matrix is aligned with the seven evidence domains used in the review and with the reconciled source-level coding in Supplementary Files 6 and 9.

Evidence Base Summary

Item	Count
Included sources	68
Reconciled domain assignments	96
Evidence domains	7
Single-domain sources	39
Two-domain sources	21
Three-domain sources	4
High-integration sources	4

Reconciled Domain Counts

Domain code	Evidence domain	Sources coded
D1	Biological interaction between malaria and red blood cell polymorphisms	20
D2	Spatial co-risk and malaria-HbS/SCD geography	10

D3	Clinical convergence and under-recognition	10
D4	Service-mediated visibility and routine data	11
D5	Structural vulnerability and access to essential SCD services	20
D6	Programme response and screening-to-care systems	19
D7	Use of syndemic theory in malaria-SCD literature	6

Table 1. Evidence Gap Matrix by Domain

Domain	What the evidence shows	Main evidence gap	Why the gap matters	Priority for future research
D1 Biological interaction	Strong evidence shows HbAS protection against severe malaria and broader malaria-relevant red cell variation involving HbC, alpha-thalassaemia, G6PD deficiency,	Few studies connect biological polymorphism evidence to contemporary SCD service planning, screening prioritisation, or clinical interpretation among children with SCA.	Biological evidence explains why HbS persists but does not by itself show where affected children are diagnosed, treated, referred, or retained in care.	Link red cell polymorphism data with newborn screening, SCA burden, malaria exposure, and clinical outcome datasets.

	ABO, Dantu, and epistasis.			
D2 Spatial co-risk	Malaria exposure, HbS frequency, and SCA birth burden are spatially heterogeneous across SSA.	Few studies integrate malaria risk, HbS or SCD burden, child population need, facility catchments, geographic access, and service readiness in one model.	Programme expansion may target inherited risk or malaria burden without knowing whether services can absorb screen-positive infants.	Develop geospatial co-risk models that combine malaria exposure, HbS/SCD burden, population need, facility readiness, travel time, and referral networks.
D3 Clinical convergence	Severe anaemia is the main clinical bridge between malaria and SCA. SCA may be hidden among children admitted with severe anaemia.	Limited evidence distinguishes malaria infection from malaria-attributable severe illness among children with SCA.	In malaria-endemic settings, malaria positivity may lead clinicians to miss underlying SCA or misattribute severe anaemia.	Conduct genotype-informed studies of severe anaemia, malaria biomarkers, parasite density, transfusion need, and outcomes among children with SCA.
D4 Service-mediated visibility	Screening, hospital genotyping, registries, and	Routine SCD data systems remain less mature than	Areas with weak diagnosis and reporting may appear to have	Build linked SCD data systems that connect

	mortality attribution make hidden SCD burden visible.	malaria routine-data systems, and recorded SCD burden may reflect service visibility rather than true need.	low SCD burden even when underlying risk is high.	screening, confirmation, clinic enrolment, admissions, transfusion, mortality, and follow-up.
D5 Structural vulnerability	Access to prophylaxis, vaccination, hydroxyurea, transfusion, referral, and follow-up shapes SCD morbidity and mortality.	Few studies measure facility readiness for essential SCD care across decentralised African systems.	Screening expansion without service readiness can produce diagnosis without treatment continuity.	Assess facility readiness for confirmatory diagnosis, hydroxyurea initiation, laboratory monitoring, transfusion, referral, and retention.
D6 Programme response	Newborn screening, point-of-care testing, implementation networks, and screening-to-care systems are emerging across SSA.	Complete cascade outcomes are inconsistently reported, especially confirmation, caregiver notification, counselling, enrolment,	Screening benefit depends on what happens after testing. Incomplete cascade data limit programme accountability.	Evaluate newborn screening-to-care cascades using standard indicators from screening through long-term retention and child survival.

		retention, and survival.		
D7 Syndemic theory	Syndemic theory helps distinguish co-occurrence from interaction, structural amplification, and excess burden.	Empirical malaria-SCD studies rarely use syndemic theory explicitly or test syndemic mechanisms.	Without explicit theory, malaria-SCD evidence may remain fragmented across disciplines.	Apply bounded syndemic frameworks that specify biological, clinical, spatial, service, structural, and programme pathways.

Table 2. Cross-Domain Evidence Gaps

Cross-domain gap	Domains involved	Current evidence pattern	What remains unresolved	Suggested study design
Malaria-HbS co-risk and service readiness are rarely integrated	D1, D2, D5, D6	HbS and malaria geography are mapped, and SCD care needs are described separately.	Which high co-risk catchments also lack confirmatory diagnosis, hydroxyurea, transfusion readiness, and referral capacity?	Geospatial service-readiness study linked to malaria and HbS/SCD burden maps.

Severe anaemia admissions may conceal SCA	D3, D4, D5	Hospital genotyping evidence shows previously unknown SCA among severely anaemic children.	How common is hidden SCA across different SSA hospital levels and malaria transmission settings?	Multicountry hospital-based severe anaemia study with genotype testing and malaria biomarkers.
Routine SCD burden may represent service visibility	D2, D4, D6	Screening and registries improve visibility, but routine SCD reporting remains fragmented.	How much SCD burden is missed in areas without newborn screening, confirmatory testing, or SCD clinics?	Linkage study using screening records, hospital records, HMIS data, and mortality data.
Screening without care continuity remains a risk	D4, D5, D6	Newborn screening is feasible, but cascade outcomes are inconsistently reported.	What proportion of screen-positive infants receive confirmatory testing, counselling, prophylaxis, hydroxyurea assessment, and long-term follow-up?	Prospective implementation cohort with cascade indicators.
Malaria-positive illness in SCA	D1, D3, D5	HbAS protects against severe	How should clinicians	Clinical diagnostic study

needs genotype-sensitive interpretation		malaria, but children with SCA may present with malaria-positive severe anaemia.	interpret malaria tests, parasite density, and PfHRP2 among children with SCA and severe anaemia?	comparing HbSS, HbAS, and HbAA children with severe anaemia and malaria positivity.
Mortality attribution underestimates SCD burden	D4, D5	SCD-attributable mortality exceeds cause-specific SCD mortality.	How often are SCD-related deaths recorded as severe anaemia, malaria, infection, respiratory illness, or other proximate causes?	Verbal autopsy and facility mortality audit linked to SCD diagnosis and screening history.
Syndemic theory is rarely operationalised empirically	D7 plus D1 to D6	Theory supports a syndemic frame, but empirical studies rarely test interaction across domains.	What analytic criteria should define malaria-SCD syndemic evidence in SSA?	Mixed-methods or realist-informed syndemic study combining biological, clinical, spatial, and service data.

Table 3. Evidence Gaps by Health-System Function

Health-system function	Evidence currently available	Gap identified	Priority indicator
Population risk identification	HbS mapping, malaria mapping, newborn screening, national surveillance in selected settings	Limited integrated co-risk and child population need mapping	Number and proportion of high co-risk catchments with active newborn screening
Screening	Newborn screening feasibility shown in several African settings	Screening coverage and geographic equity often not reported consistently	Percentage of newborns screened among eligible births by catchment
Confirmatory diagnosis	Confirmatory testing recognised as necessary	Confirmatory testing completion often underreported	Percentage of screen-positive infants receiving confirmatory testing
Result return	Some programme reports describe tracking and notification	Time to result return and caregiver notification inconsistently reported	Median time from screening to caregiver notification
Counselling	Identified as part of screening-to-care systems	Counselling completion often not reported	Percentage of confirmed infants whose caregivers receive counselling
Clinic enrolment	Some programmes report enrolment	Enrolment into SCD care often incompletely tracked	Percentage of confirmed infants enrolled in SCD clinic care

Preventive care	Penicillin and vaccination are recognised as essential	Initiation and continuity data are limited in many settings	Percentage initiated on prophylaxis and vaccination plan
Hydroxyurea	SSA trials show feasibility and benefit	Readiness, initiation, monitoring, and retention remain unevenly assessed	Percentage eligible assessed and initiated on hydroxyurea
Transfusion readiness	Severe anaemia trials show transfusion relevance	Facility readiness for safe transfusion is not consistently mapped with SCD burden	Facilities with blood availability, cross-match capacity, and referral pathways
Referral systems	Hub-and-spoke logic is implied in programme evidence	Referral completion and feedback loops are rarely measured	Percentage completing referral after positive screen or severe admission
Retention	Follow-up is repeatedly identified as important	Long-term retention is inconsistently reported	Retention at 6, 12, 24, and 60 months after diagnosis
Mortality ascertainment	Mortality attribution studies show undercounting	Linkage between SCD diagnosis and death reporting is limited	Deaths among diagnosed children with cause and contributing condition review

Table 4. Priority Research Questions

Priority area	Research question	Linked domains
Spatial planning	Where do malaria exposure, HbS/SCD burden, child	D2, D4, D5, D6

	population need, weak service readiness, and low routine visibility overlap?	
Severe anaemia	What proportion of children admitted with severe anaemia have previously unknown SCA across different malaria transmission settings?	D3, D4
Clinical attribution	How should malaria-positive severe anaemia be interpreted among children with SCA compared with HbAA and HbAS children?	D1, D3, D5
Screening coverage	Which catchments are reached or missed by newborn screening programmes, and how does this compare with estimated SCA birth burden?	D2, D4, D6
Screening-to-care linkage	What proportion of screen-positive infants complete confirmation, notification, counselling, clinic enrolment, prophylaxis, hydroxyurea assessment, and follow-up?	D4, D5, D6
Facility readiness	Which facilities can provide essential SCD care, including confirmatory testing, prophylaxis, hydroxyurea,	D5, D6

	laboratory monitoring, transfusion, and referral?	
Routine data	How accurately do routine health information systems capture SCD diagnoses, admissions, transfusions, deaths, and follow-up?	D4, D5
Mortality attribution	How much SCD-attributable mortality is recorded under malaria, severe anaemia, infection, respiratory illness, or other proximate causes?	D3, D4, D5
Syndemic measurement	What minimum evidence is required to classify malaria-SCD relationships as co-occurrence, interaction, convergence, structural amplification, or programme-mediated risk?	D7 plus D1 to D6
Implementation strategy	Which hub-and-spoke or decentralised screening-to-care models produce the best retention and survival outcomes in high co-risk areas?	D4, D5, D6

Table 5. Suggested Core Indicators for Future Malaria-SCD Syndemic Studies

Indicator category	Suggested indicator	Rationale
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Malaria exposure	Parasite prevalence, malaria incidence, malaria mortality, or transmission intensity by catchment	Measures ecological exposure relevant to HbS selection and current co-risk
HbS/SCD burden	HbAS prevalence, HbSS prevalence, estimated SCA births, confirmed SCD diagnoses	Measures inherited sickle-related risk and burden
Child population need	Births, under-five population, paediatric admissions, severe anaemia admissions	Identifies population denominator for planning
Clinical convergence	Severe anaemia admissions, malaria positivity, parasite density, PfHRP2, transfusion need, respiratory distress	Captures malaria-SCD clinical overlap
Under-recognition	Previously unknown SCA among hospital admissions or severe anaemia cases	Measures hidden burden in clinical settings
Screening cascade	Screened, positive screen, confirmatory test, caregiver notified, counselled, enrolled	Measures movement from testing to care entry
Care cascade	Prophylaxis, vaccination, hydroxyurea assessment, hydroxyurea initiation, transfusion access, referral completion	Measures continuity of essential care
Retention	Follow-up attendance at 6, 12, 24, and 60 months	Measures continuity after diagnosis

Routine visibility	SCD-coded outpatient visits, admissions, transfusions, deaths, registry entries	Measures health-system visibility
Mortality attribution	Cause-specific death, SCD-attributable death, contributing causes	Measures under-attribution and hidden mortality
Service readiness	Diagnostic capacity, hydroxyurea availability, laboratory monitoring, blood availability, specialist referral	Measures capacity to act on screening
Geographic access	Travel time to screening site, confirmatory laboratory, SCD clinic, transfusion facility	Measures spatial access barriers

Table 6. Research Priority Ranking

Rank	Priority	Justification	Suggested timeframe
1	Integrated malaria-HbS/SCD co-risk and service readiness mapping	This is the most direct planning gap for risk-informed newborn screening and essential care expansion.	Immediate
2	Severe anaemia genotyping and malaria biomarker studies	This directly addresses clinical under-recognition and diagnostic attribution among hospitalised children.	Immediate

3	Screening-to-care cascade evaluation	Screening benefit depends on confirmation, notification, counselling, enrolment, treatment, referral, and retention.	Immediate
4	Facility readiness assessment for essential SCD care	Screening expansion requires facilities capable of providing or referring for essential care.	Immediate to medium term
5	Routine SCD data strengthening and registry linkage	Routine data are needed for monitoring burden, care continuity, admissions, transfusion, and mortality.	Medium term
6	Mortality attribution studies	SCD-attributable mortality is likely underestimated when deaths are assigned to proximate causes.	Medium term
7	Geographic access and referral network studies	Catchment planning requires travel time, referral completion,	Medium term

		and hub-and-spoke modelling.	
8	Empirical syndemic modelling	Stronger syndemic evidence requires explicit analysis of interaction, structural amplification, and programme mediation.	Medium to long term

Table 7. Recommended Study Designs

Research need	Recommended design	Key data sources
Identify high co-risk areas	Geospatial modelling	Malaria maps, HbS/SCD prevalence, births, child population, facility locations, travel time
Detect hidden SCA in severe anaemia	Hospital-based cross-sectional or cohort study	Severe anaemia admissions, genotype testing, malaria diagnostics, PfHRP2, transfusion records
Evaluate screening-to-care cascade	Prospective implementation cohort	Newborn screening database, confirmatory testing, clinic enrolment, care records
Measure facility readiness	Health facility assessment	Laboratory capacity, hydroxyurea stock, blood bank readiness, referral protocols, staffing

Assess routine visibility	Data linkage study	HMIS, hospital records, SCD clinic registers, screening records, mortality records
Estimate SCD-attributable mortality	Verbal autopsy and mortality audit	Death certificates, verbal autopsy, clinic history, screening records
Understand implementation barriers	Mixed-methods implementation study	Caregiver interviews, provider interviews, programme records, facility assessments
Test syndemic pathways	Realist-informed or mixed-methods study	Biological, spatial, clinical, routine data, service readiness, and qualitative data

Table 8. Programme Planning Implications Derived From Evidence Gaps

Planning problem	Evidence gap behind the problem	Planning implication
Where should newborn screening expand first?	Limited integrated co-risk and service readiness mapping	Prioritise catchments using combined malaria exposure, HbS/SCD burden, birth volume, service readiness, and access indicators.
Which facilities should be strengthened?	Facility readiness is rarely mapped against sickle burden	Strengthen facilities in high-burden catchments with capacity gaps in diagnosis, hydroxyurea, transfusion, and referral.

How should severe anaemia admissions be interpreted?	SCA under-recognition among severe anaemia admissions is under-measured	Integrate SCD testing into severe anaemia pathways in high-risk settings.
How should routine SCD data be used?	Routine data may reflect service visibility rather than true burden	Interpret recorded SCD burden alongside screening coverage, diagnostic access, and clinic availability.
How should screening success be measured?	Screening alone does not equal care continuity	Monitor the full cascade from screening to long-term retention and survival.
How should syndemic evidence be used?	Empirical syndemic modelling is limited	Use a bounded framework that separates co-occurrence, biological interaction, clinical convergence, structural vulnerability, and programme response.